MONTH 1: PYTHON BASICS AND CORE CONCEPTS

- Week 1: Python Syntax and Data Types
- Tasks:
  - Learn basic Python syntax, variables, and data types (strings, lists, dictionaries).
- Project: Create a simple calculator application.
- Week 2: Control Flow and Functions
- Tasks:
  - Understand loops, conditionals, and functions in Python.
- Project: Develop a quiz application that uses functions to evaluate answers.
- Week 3: File Handling and Exception Handling
- Tasks:
  - Learn to read/write files and handle exceptions.
- Project: Create a contact management system that stores contacts in a file.
- Week 4: Object-Oriented Programming (OOP)
- Tasks:
  - Understand the principles of OOP in Python (classes, objects, inheritance).
- Project: Develop a simple banking system using OOP concepts.

MONTH 2: INTERMEDIATE PYTHON AND LIBRARIES

- Week 5: Working with Libraries (NumPy, Pandas)
- Tasks:
  - Learn to manipulate data using NumPy and Pandas.
- Project: Analyze a dataset (e.g., CSV file) and perform basic data operations.
- Week 6: Web Scraping with BeautifulSoup
- Tasks:
  - Learn web scraping techniques using BeautifulSoup.
- Project: Build a web scraper to extract data from a website and store it in a file or database.
- Week 7: APIs and JSON
- Tasks:
  - Learn to work with REST APIs and JSON data.
- Project: Create a weather application that fetches data from an API and displays it to the user.
- Week 8: Data Visualization (Matplotlib, Seaborn)
- Tasks:
  - Learn to create plots and visualizations using Matplotlib and Seaborn.
- Project: Visualize the data from Week 5's project with various charts.

MONTH 3: ADVANCED TOPICS AND PROJECT

- Week 9: Database Interaction (SQLite, MySQL)
- Tasks:
  - Learn to interact with databases using Python.
- Project: Build a simple inventory management system with a database backend.
- Week 10: Automation with Python
- Tasks:
  - Learn to automate tasks with Python (e.g., file operations, web interactions).
- Project: Create a script to automate the backup of files or emails.
- Week 11: Testing and Debugging
- Tasks:
  - Understand testing methodologies (unit tests) and debugging techniques.
- Project: Write test cases for one of the previous projects and improve code quality.
- Week 12: Final Project & Presentation
- Project: Develop a final project of your choice, integrating multiple libraries and techniques learned. Present the project with documentation and a live demo.